



Customer Behavior Analysis Report

An in-depth analysis of **5,000+ purchases** across multiple product categories — uncovering spending patterns, purchasing trends, and actionable business insights to optimize marketing and maximize revenue.

Project Objectives



Purchasing Behavior

Analyze how customers shop and spend



Top Performers

Identify high-performing categories and products



Discount Impact

Evaluate how discounts affect sales



Segmentation

Segment customers by behavior



Actionable Insights

Deliver data-driven recommendations

Dataset at a Glance

5000+

Total Customers

30

Unique Items

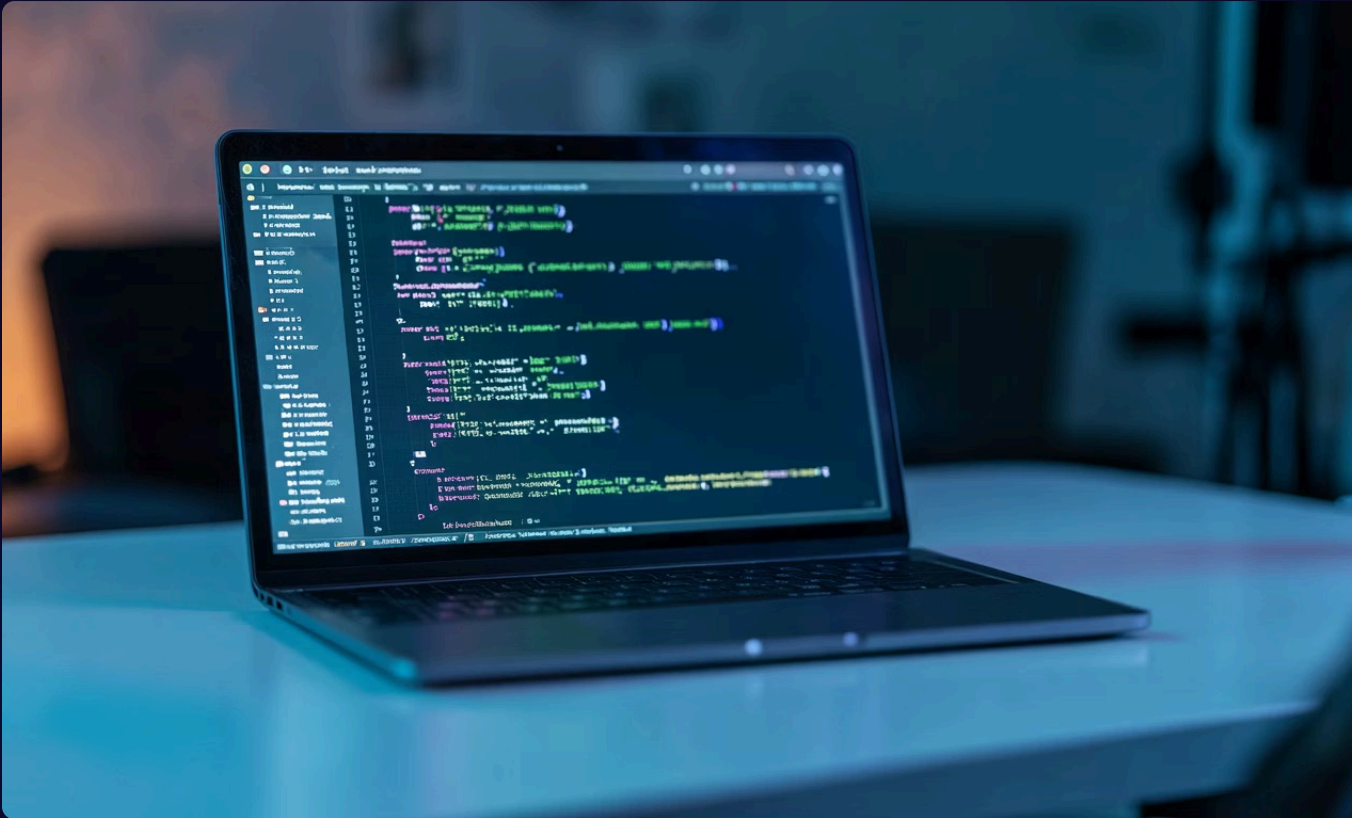
3.62

Avg Review Rating

\$198.84

Avg Purchase Amount

The dataset includes 12 attributes per transaction: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount, Review Rating, Payment Method, Shipping Type, Subscription Status, Previous Purchases, and Location.



DATA PREPARATION

Exploratory Data Analysis

The dataset was cleaned and prepared using Python before analysis. Key steps included:

→ Missing Values

Handled nulls with `isnull()`

→ Data Types

Verified with `df.info()`

→ Statistical Summary

Generated via `df.describe()`

→ Formatting

Rounded review ratings for consistency

SQL ANALYSIS

Revenue by Category

Electronics

\$486,367 — Highest revenue category

Accessories

\$194,867 — Second place

Clothing

\$183,188 — Third place

Footwear

\$111,236 — Fourth place

Outerwear

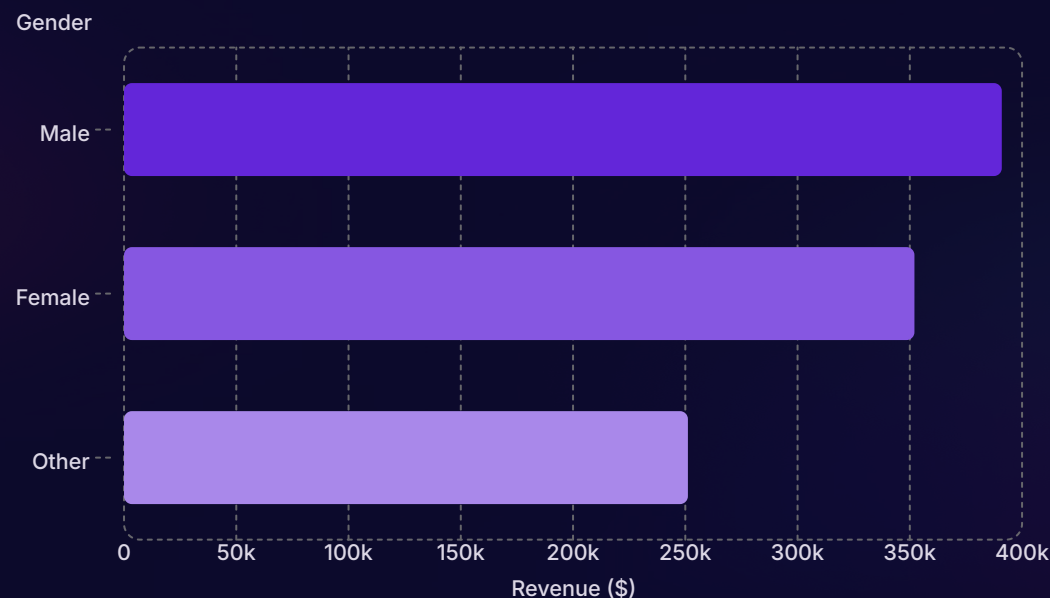
\$18,524 — Lowest revenue

Discounts & Customer Spending

Do Discounts Increase Purchase Value?



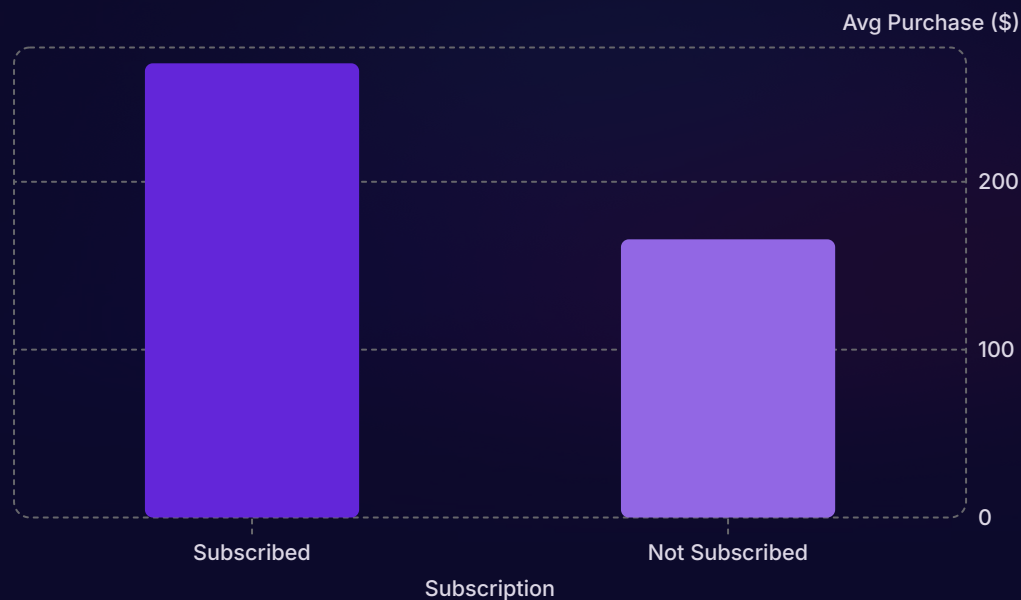
Revenue by Gender



Customers using discounts spend **\$219.22** on average vs. **\$182.24** without — confirming discounts drive higher purchase values. Male customers contribute the largest share of total revenue.

Subscriptions & Customer Segments

Subscribers Spend More



Customer Segments

Subscribers spend **63% more** per transaction. Loyal customers represent **62%** of the customer base — a key retention target.

Loyal Customers

3,085 — highest segment

Returning Customers

1,354

New Customers

561

POWER BI

Interactive Dashboard

A Power BI dashboard visualizes five key metrics and charts covering revenue by location, gender, shipping type, category, and payment method – with filters for category, gender, and subscription status.

5K Customers

30 Unique Items

3.62 Avg Review

\$198.84 Avg Amount

\$994.18K Total Revenue

KEY FINDINGS

Key Insights

Electronics Dominates

Highest-revenue category at \$486K, led by Phone, Headphones, and Watch

Male Customers Lead

Largest revenue share; Debit Card is the top payment method

Discounts Drive Value

Discounted purchases average \$219 vs. \$182 without

Loyal Customers Matter

3,085 loyal customers contribute significantly to total revenue

Location Variance

Phoenix and Los Angeles outperform New York in revenue

RECOMMENDATIONS

Business Recommendations



Focus on Electronics

Concentrate marketing on the highest-revenue category



Optimize Discounts

Refine discount strategies to maintain profitability while driving volume



Reward Loyal Customers

Target loyal customers with exclusive offers to maximize retention



Promote Subscriptions

Encourage subscription plans — subscribers spend 63% more per order